Sanskar Jain

Backend Engineer | Python, Java, SQL | Infrastructure at Scale
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Summary

Backend engineer with 2+ years of experience building infrastructure at scale, designing high-throughput data pipelines, and optimising SQL-heavy systems at Verisk Analytics. Proficient in Python, Java, and SQL with a strong grasp of data structures, algorithms, and system performance metrics. Proven ability to ship scalable, reliable services handling 200K+ weekly requests in production.

Key Skills

Languages: Python, Java, SQL, C#, TypeScript
Infrastructure & Scale: Distributed Systems, Microservices, Event-Driven Architecture, Kafka, Redis, AWS (S3, SNS, SQS, EKS, Lambda)
Databases & SQL: PostgreSQL, MySQL, MongoDB, Query Optimisation, Indexing, Connection Pooling
CS Fundamentals: Data Structures & Algorithms, Time/Space Complexity Analysis, System Design
Tools & DevOps: Docker, Kubernetes, Terraform, CI/CD, Splunk, Git, Jira

Experience

Verisk Analytics Jan 2024 – Present
Software Engineer II Hyderabad, India

- Architected **RulesRunner**, a distributed microservice processing **200K+** weekly messages via **SQS**, reducing client-side latency from 3000ms to 500ms (**83% reduction**) through efficient queue-based decoupling and algorithm-optimised rule evaluation.
- Implemented **high-performance SQL queries** and indexing strategies on PostgreSQL/MongoDB, improving API response time by **34%** via Redis caching, connection pooling, and targeted query optimisation on hot-path endpoints.
- Engineered scalable data pipeline for import/export of **30+ data types** using **Python** scripts, async processing, AWS S3 presigned URLs, and MongoDB, reducing data transfer time by **38%** and user-reported issues by **25%**.
- Built production **REST APIs** handling **200K+ requests/week** with structured error handling, input validation, and O(1) lookup patterns for low-latency responses.
- Developed **Python** automation for data validation, log analysis, and performance benchmarking, reducing manual effort by **15+ hours/week** and enabling data-driven decision-making.
- Cut technical debt by **55%** and improved runtime performance by **30%** through service modernisation, optimising time-complexity of core algorithms and memory allocation patterns.
- Instrumented services with structured logging and **Splunk** dashboards tracking latency percentiles, throughput, and error rates to proactively identify bottlenecks.

Projects

Expense Tracker – Full-Stack Microservices Aug 2023 – Dec 2023
Java, Spring Boot, Kafka, PostgreSQL, Docker

- Built expense analytics platform using **Java**/Spring Boot with **5+ microservices**, async **Kafka** event streaming, and **SQL**-based transaction aggregation with optimised joins.
- Designed schema with indexing and normalisation; wrote complex **SQL queries** for reporting and analytics across **1000+** simulated transactions; containerised with Docker.

Stock Recommendation & Prediction Apr 2022 – Jun 2022
Python, MySQL, Data Analysis, LSTM

- Developed **Python**-based stock advisory platform using LSTM models for 30-day price predictions with **MySQL** backend for historical data storage and retrieval.
- Wrote optimised **SQL queries** for time-series data aggregation and implemented efficient data structures for real-time price lookups.

Certifications

AWS Certified Cloud Practitioner Amazon Web Services | Oct 2024

Education

B.Tech in Computer Engineering, Vishwakarma Institute of Technology, Pune (S.P.P.U) 2024 | CGPA: 8.8